

In this section we deal with two results—Hensel's lemma and the Weierstrass preparation theorem—that show how far one can go in simplifying power series $F(x, y)$ if we do not want to pass to Puiseux series.

Let us start with the simplest case, which is also easy to derive from Newton's method.

LEMMA 1.84. *Let $F(x, y) \in K[[x]][y]$ be a polynomial in y whose coefficients are power series in x . Assume that α is a simple root of $F(0, y) = 0$. Then $F(x, y) = 0$ has a unique power series solution*

$$y = \sum_{i=0}^{\infty} b_i x^i \quad \text{with } b_0 = \alpha.$$

Proof. We try to find the coefficients of the assumed power series solution $y = b_0 + b_1x + b_2x^2 + \cdots$ inductively.

That is, first, we solve modulo x ; this amounts to solving the ordinary polynomial equation

$$F(0, y) = 0 \quad \text{or, equivalently, } F(x, y) \equiv 0 \pmod{x}.$$

By assumption, $b_0 = \alpha$ is a solution. Then we try to get a linear term b_1x such that

$$F(x, b_0 + b_1x) \equiv 0 \pmod{x^2}.$$

Next comes the quadratic part b_2x^2 and so on.

Assume that we already found $b_0 + b_1x + \cdots + b_mx^m$ such that

$$F(x, b_0 + b_1x + \cdots + b_mx^m) \equiv 0 \pmod{x^{m+1}}.$$

Next we want to get b_{m+1} such that

$$F(x, b_0 + b_1x + \cdots + b_mx^m + b_{m+1}x^{m+1}) \equiv 0 \pmod{x^{m+2}}.$$

Set $y_m(x) := b_0 + b_1x + \cdots + b_mx^m$. By Taylor's theorem

$$F(x, y_m(x) + b_{m+1}x^{m+1}) \equiv F(x, y_m(x)) + \frac{\partial F}{\partial y}(x, y_m(x))b_{m+1}x^{m+1},$$

modulo $(x^{2(m+1)})$. By assumption

$$F(x, y_m(x)) \equiv C_{m+1}x^{m+1} \pmod{x^{m+2}}$$

for some constant C_{m+1} and

$$\frac{\partial F}{\partial y}(x, y_m(x))x^{m+1} \equiv \frac{\partial F}{\partial y}(0, b_0)x^{m+1} \pmod{x^{m+2}}.$$

Thus we need to solve the equation

$$C_{m+1} + \frac{\partial F}{\partial y}(0, b_0)b_{m+1} = 0.$$

1.11. Hensel's lemma and the Weierstrass preparation theorem

As we saw already in Newton's method, resolving curve singularities is very closely related to solving algebraic equations $f(x, y) = 0$. In general there are no power series solutions, only Puiseux series solutions.

This has a unique solution if $\partial F/\partial y(0, b_0) \neq 0$, that is, when b_0 is a simple root of $F(0, y) = 0$. Thus inductively we find a unique power series solution

$$y = \sum_{i=0}^{\infty} b_i x^i.$$

If, however, $\partial F/\partial y(0, b_0) = 0$, that is, when b_0 is a multiple root of $F(0, y) = 0$, then we have no solution in general. $\square$

REMARK 1.85. The Newton iteration method for finding roots of a real or complex polynomial $f(x)$ is the rule

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}.$$

One should recognize the method of (1.84) as following this iteration and truncating at each step. We work in a ring $K[[x]]$ instead of a field, and hence the invertibility of $f'(x_i)$ is a bigger problem than usual.

If $F(0, y)$ has only multiple roots, we have no power series solutions in general, but we can separate the different roots of $F(0, y)$ from each other without passing to Puiseux series. This is the content of Hensel's lemma.

THEOREM 1.86 (Hensel's lemma). *Let $F(x, y) \in K[[x]][y]$ be a polynomial in y whose coefficients are power series in x . Assume that*

$$F(0, y) = g(y)h(y),$$

where g and h are relatively prime. Then we can write

$$F(x, y) = G(x, y)H(x, y), \quad \text{where } G(0, y) = g(y) \text{ and } H(0, y) = h(y).$$

Moreover, there is a unique solution G, H such that

$$\deg_y(G(x, y) - g(y)) < \deg_y g(y).$$

Proof. As before, we try to find inductively

$$G_m(x, y) := \sum_{i=0}^m g_i(y)x^i \quad \text{and} \quad H_m(x, y) = \sum_{i=0}^m h_i(y)x^i$$

such that

$$G_m(x, y)H_m(x, y) \equiv F(x, y) \pmod{(x^{m+1})}.$$

The starting case is $G_0(x, y) = g(y)$ and $H_0(x, y) = h(y)$. In order to go from m to $m+1$, we need to solve, modulo (x^{m+2}) , the equation

$$(G_m(x, y) + g_{m+1}(y)x^{m+1})(H_m(x, y) + h_{m+1}(y)x^{m+1}) \equiv F(x, y).$$

By assumption,

$$G_m(x, y)H_m(x, y) \equiv F(x, y) + f_{m+1}(y)x^{m+1} \pmod{(x^{m+1})}$$

for some $f_{m+1}(y)$, and thus we are down to solving

$$g(y)h_{m+1}(y) + h(y)g_{m+1}(y) = f_{m+1}(y). \quad (1.86.1)$$

Since g and h are relatively prime, $g(y)a(y) + h(y)b(y) = 1$ is solvable, so we can take $h_{m+1}(y) = a(y)f_{m+1}(y)$ and $g_{m+1}(y) = b(y)f_{m+1}(y)$.

Moreover, if $u(y)g(y) + v(y)h(y) = f_{m+1}(y)$ is any solution, then we can uniquely write $v(y) = g(y)s(y) + g_{m+1}(y)$, where $\deg g_{m+1} < \deg g$. This gives a unique solution of (1.86.1) with $\deg g_{m+1} < \deg g$ for every $m \geq 1$. In this case (1.86.1) implies that $\deg_y H \leq \deg_y F - \deg_y g$; thus H is also a polynomial in y .

Thus there is a unique solution where $\deg_y(G(x, y) - g(y)) < \deg_y g(y)$. $\square$

It is worthwhile to look at the above proof to see if one can replace the polynomial ring $K[[x]][y]$ by some other ring. The first observation is that we used only that $K[[x]][y]/(x^m)$ is a polynomial ring in y , and so we could have worked in the ring $K[y][[x]]$ to start with. (Note that this is indeed a bigger ring. For instance, $\sum x^i y^i \in K[y][[x]] \cap K[x][[y]]$, but it is not in $K[[x]][y]$.) Without any change in the proof, we get the following general form.

PROPOSITION 1.87 (Abstract Hensel lemma). *Let R be a ring and $F(x) := \sum_{i=0}^{\infty} r_i x^i$ a power series in x with coefficients in R . Assume that*

$$r_0 = g_0 h_0 \quad \text{and} \quad g_0 R + h_0 R = R.$$

Then we can write

$$F(x) = G(x)H(x), \quad \text{where } G(0) = g_0 \text{ and } H(0) = h_0.$$

Moreover, if we fix representatives r^* for every residue class $R/(g_0)$, then there is a unique solution where $G(x) = g_0 + \sum_{i \geq 1} g_i^* x^i$. $\square$

As a simple consequence, we obtain the Weierstrass preparation theorem, which shows that any power series $F(x, y)$ can be made into a polynomial in one of the variables.

COROLLARY 1.88 (Weierstrass preparation theorem). *Let $F(x, y) \in K[[x, y]] = K[[y]][[x]]$ be a power series. Assume that y^n appears in F with nonzero coefficient and n is the smallest such exponent. Then one can write $F(x, y)$ uniquely as*

$$F(x, y) = (y^n + g_{n-1}(x)y^{n-1} + \cdots + g_0(x)) \cdot (\text{unit}),$$

where $g_i \in K[[x]]$.

Proof. We apply (1.87) to $R = K[[y]]$. Here we can write $F(0, y) = y^n \cdot u(y)$, where $u(y)$ is a unit; set $g_0 := y^n$ and $h_0 := u$. The unit h_0 alone generates $K[[y]]$, and every $\phi(y) \in K[[y]]$ can be written uniquely as

$\phi(y) = y^n \cdot s(y) + u(y) \cdot t(y)$, where $\deg t < n$. Indeed, $t(y)$ is the sum of the first n terms of $\phi(y) \cdot u(y)^{-1}$. $\square$

1.89 (Embedded resolution using Weierstrass' theorem). Let $0 \in C$ be a singular point of a curve on a smooth surface S . If $\text{mult}_0 C = m$, then we can choose local coordinates such that y^m appears in the equation of C with nonzero coefficient. Thus by (1.88) in the completion $\hat{\mathcal{O}}_{0,S}$ we can write the equation as

$$\hat{C} = (y^m + g_{m-1}(x)y^{m-1} + \cdots + g_0(x) = 0).$$

If the characteristic does not divide m , then by a change of coordinates $y \mapsto y - \frac{1}{m}g_{m-1}$ we get the simpler equation

$$\hat{C} = (y^m + a_{m-2}(x)y^{m-2} + a_{m-3}(x)y^{m-3} + \cdots + a_0(x) = 0). \quad (1.89.1)$$

This means that $(y = 0)$ is a curve of maximal contact (1.80), and the computation of the blow-ups is especially easy in this case.

The assumption $\text{mult}_0 C = m$ means that $\text{mult}_0 a_{m-j} \geq j$. On the first blow-up everything happens on the chart with coordinates $y_1 = y/x, x_1 = x$. The birational transform of $\hat{C}$ is

$$\hat{C}_1 = (y_1^m + x_1^{-2}a_{m-2}(x_1)y_1^{m-2} + x_1^{-3}a_{m-3}(x_1)y_1^{m-3} + \cdots + x_1^{-m}a_0(x_1) = 0).$$

The form of the equation is unchanged, and $\text{mult}_0 \hat{C}_1 \geq m$ iff $\text{mult}_0 a_{m-j} \geq 2j$ for every j . Thus after k blow-ups we have coordinates $y_k = y/x^k, x_k = x$ and the birational transform $\hat{C}_k$ of $\hat{C}$ is given by the equation

$$(y_k^m + x_k^{-2k}a_{m-2}(x_k)y_k^{m-2} + x_k^{-3k}a_{m-3}(x_k)y_k^{m-3} + \cdots + x_k^{-mk}a_0(x_k) = 0).$$

From this we conclude that

$$\text{mult}_0 \hat{C}_k = m \quad \text{iff} \quad k \leq \min \left\{ \frac{\text{mult}_0 a_{m-i}(x)}{i} : i = 2, \dots, m \right\}.$$

Thus we can read off the precise number of blow-ups needed to lower the multiplicity from the equation (1.89.1).

This is the simplest illustration of the general theory of coefficient ideals, which is used in the proof of resolution in higher dimensions.

In order to make this approach into a proof of resolutions, one needs to analyze the relationship between blowing up the completion $\hat{\mathcal{O}}_{0,S}$ and blowing up S . Everything works out as expected, and the easy details are left to the reader.

ASIDE 1.90 (Weighted blow-ups). One can achieve the first k steps of the above blow-up sequence by just one *weighted blow-up*. In this case this amounts to blowing up the ideal (y, x^k) . The resulting surface has a singular point, but the birational transform of the curve does not pass through it, so it does not interfere with the resolution of the curve.

Weighted blow-ups and more generally toric methods (see [Ful93]) work very efficiently if we know beforehand the optimal choice of the coordinate system or if the equation has only a few monomials. See [KSC04, Sec.6.5] for a nice example and for complete definitions.

The proof of the following form of Hensel's lemma is also left to the reader.

PROPOSITION 1.91. For $F(x, y) \in K[[x]][y]$, let $y_{\text{app}}(x) \in K[[x]]$ be an approximate solution; that is,

$$F(x, y_{\text{app}}(x)) \equiv 0 \pmod{(x^{m+1})}$$

for some m . Assume that

$$m \geq 2n, \quad \text{where} \quad n := \text{mult}_x \frac{\partial F}{\partial y}(x, y_{\text{app}}(x)).$$

Then there is a solution $y(x)$ such that

$$y(x) \equiv y_{\text{app}}(x) \pmod{(x^{m+1-n})}. \quad \square$$

It is straightforward to generalize the results of this section to the case where instead of one variable x we have several variables $x_1, \dots, x_n$.

THEOREM 1.92 (General Hensel lemma). Let R be a ring and

$$F(x_1, \dots, x_n) := \sum_{i_1, \dots, i_n} r_{i_1, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n}$$

a power series in $x_1, \dots, x_n$ with coefficients in R . Assume that

$$r_{0, \dots, 0} = gh \quad \text{and} \quad gR + hR = R.$$

Then we can write

$$F(\mathbf{x}) = G(\mathbf{x})H(\mathbf{x}), \quad \text{where } G(0, \dots, 0) = g \text{ and } H(0, \dots, 0) = h.$$

Moreover, if we fix representatives r^* for every residue class $R/(g)$, then there is a unique solution, where $g_{i_1, \dots, i_n} = g_{i_1, \dots, i_n}^*$ whenever $(i_1, \dots, i_n) \neq (0, \dots, 0)$. $\square$

As before, applying this to $K[[y]][[x_1, \dots, x_n]]$ we get the following.

THEOREM 1.93 (Weierstrass preparation theorem). Let K be a field and

$$F(x_1, \dots, x_n, y) \in K[[x_1, \dots, x_n, y]]$$

a power series. Assume that y^m appears in F with nonzero coefficient and m is the smallest such. Then one can write F uniquely as

$$F = (y^m + g_{m-1}(\mathbf{x})y^{m-1} + \cdots + g_0(\mathbf{x})) \cdot (\text{unit}),$$

where $g_i(\mathbf{x}) \in K[[x_1, \dots, x_n]]$. $\square$

1.12. Extensions of $K((t))$ and algebroid curves

In this section we describe finite field extensions of the field of formal Laurent series $K((t))$ over an algebraically closed field K . Equivalently, we solve equations $f(t, y) = 0$, where $f(t, y) \in K((t))[y]$.

This is the setting of Newton's method of the rotating ruler, and it is straightforward to rewrite it in a more algebraic setting.

More generally, we can work with germs of formal curves; that is, we consider $F(x, y) = 0$, where $F(x, y) \in K[[x, y]]$ is a formal power series. These are also called *algebroid curves* or *singularities*.

THEOREM 1.94. *Let K be an algebraically closed field and $K((t))$ the field of formal Laurent series. Let $L \supset K((t))$ be a finite degree field extension. Assume that $\text{char } K$ does not divide $\deg |L/K((t))|$.*

Then $L = K((t^{1/m}))$, where $m = \deg |L/K((t))|$.

Proof. First we establish that $L \subset K((t^{1/M}))$ for some M , using induction on $\deg |L/K((t))|$.

Pick any $y \in L \setminus K((t))$. It satisfies a minimal equation

$$\sum_{i=0}^n a_i(t)y^i = 0, \quad \text{where } a_i(t) \in K((t)) \text{ and } \text{char } K \nmid n.$$

We can multiply through by a large power of t to assume that $a_i(t) \in K[[t]]$. Then multiply the equation by $a_n(t)^{n-1}$, and substitute $y \mapsto a_n(t)^{-1}y$ to get a monic equation

$$y^n + b_{n-1}(t)y^{n-1} + \cdots + b_0(t) = 0, \quad \text{where } b_i(t) \in K[[t]].$$

Finally, since the characteristic does not divide n , we can substitute $y \mapsto y - \frac{1}{n}b_{n-1}(t)$ to get

$$F(t, y) := y^n + c_{n-2}(t)y^{n-2} + \cdots + c_0(t) = 0, \quad \text{where } c_i(t) \in K[[t]].$$

(This last transformation is a seemingly minor convenience, but the proof really hinges on it. We use it through the following elementary observation: A polynomial $y^n + a_{n-2}y^{n-2} + \cdots + a_0 \in K[y]$ is not an n th power of a linear polynomial if $\text{char } K \nmid n$, unless $a_{n-2} = \cdots = a_0$.)

Let us look now at the leading part in t , which is $F(0, y) \in K[y]$, and distinguish two cases.

Case 1. If $F(0, y)$ has at least two distinct roots, then it can be written as $F(0, y) = F_1(0, y)F_2(0, y)$, where F_1 and F_2 are relatively prime. Thus by the Hensel lemma (1.86), F has a corresponding nontrivial factorization $F(t, y) = F_1(t, y)F_2(t, y)$, contradicting the assumption that F is a minimal polynomial of y .

Case 2. If F has only one root, then $F(0, y) = (y - \alpha)^n$ for some $\alpha \in K$. Comparing the coefficients of y^{n-1} we see that $\alpha = 0$, and so $F(0, y) = y^n$. Thus t divides every $c_i(t)$.

If by chance $t^{r(n-i)}|c_i(t)$ for some $r > 0$ and every i , then we can substitute $y \mapsto t^r y$ and divide by t^{nr} to get the equation

$$\left(\frac{y}{t^r}\right)^n + \frac{c_{n-2}(t)}{t^{2r}}\left(\frac{y}{t^r}\right)^{n-2} + \cdots + \frac{c_0(t)}{t^{nr}} = 0, \quad \text{where } \frac{c_i(t)}{t^{r(n-i)}} \in K[[t]].$$

The previous argument with the Hensel lemma now shows that $t|c_i(t)/t^{r(n-i)}$ for every i .

We do not get much out of this if we take r to be an integer, but allowing r to be rational turns out to be very useful.

The largest value of r allowed without ending up with poles in some $c_i(t)/t^{r(n-i)}$ is

$$r = \min_i \left\{ \frac{\text{mult}_0 c_i(t)}{n-i} \right\}.$$

The minimum is achieved for some $i_0 \leq n-2$, and then $r = u/v$ with $v|n-i_0 \leq n-2$. By setting $s = t^{1/v}$, $y_1 = y/t^r$ and $d_i(t^{1/v}) = c_i(t)/t^{r(n-i)}$, we get an equation

$$G(y_1, s) := y_1^n + d_{n-2}(s)y_1^{n-2} + \cdots + d_0(s) = 0.$$

As before, the $d_i(s)$ are power series in s , and $d_{i_0}(s)$ has a nonzero constant term.

This means that $G(y_1, 0)$ is different from y_1^n . Hence by the argument in Case 1, $G(y_1, s)$ factors nontrivially over $K((s)) = K((t^{1/v}))$. (It is not hard to see that $\text{char } K \nmid v$, but let us not worry about this right now.)

In any case, at least one of the factors has a degree that is not divisible by $\text{char } K$, so by induction on the degree, we conclude that $y \in K((t^{1/M}))$ for some M . More precisely, at first we know this only for some conjugate of y , but $K((t^{1/M}))/K((t))$ is a normal extension, so in fact we get $y \in K((t^{1/M}))$.

This implies that the degree of $L(t^{1/M})/K((t^{1/M}))$ is a proper divisor of the degree of $L/K((t))$. By induction we conclude that $L \subset K((t^{1/M}))$ (for a possibly larger M).

In characteristic zero, $K((t^{1/M}))/K((t))$ is a Galois extension with cyclic Galois group, and so every intermediate field is of the form $K((t^{1/m}))$ for some $m|M$. Hence, $L = K((t^{1/m}))$, where $m = \deg |L/K((t))|$.

In positive characteristic write $M = M'p^a$, where $p = \text{char } K$ and p does not divide M' . Then $K((t^{1/M'}))$ is the unique maximal separable subextension of $K((t^{1/M}))$. Since $\text{char } K$ does not divide $\deg |L/K((t))|$, $L/K((t))$ is separable, and thus $L \subset K((t^{1/M'}))$. Now we can complete the argument as in the characteristic zero case. $\square$

By contrast, degree p extensions in characteristic p have no simple description. First of all, there is a unique inseparable extension of degree p , obtained by adjoining all p th roots.

In many problems, however, the main difficulty is caused by degree p separable extensions and *wild ramification*.

1.95 (Artin-Schreier theory). cf. [Ser79, X.3] Let k be a field of characteristic p and k'/k a degree p Galois extension. Then k' is the splitting field of a polynomial $y^p - y - a$, and $a \in k$ is unique modulo elements of the form $\{b^p - b : b \in k\}$.

Assume next that $k = K((t))$ and K is perfect. If $g \in tK[[t]]$, then

$$-g = (\sum_{i \geq 1} g^{p^i})^p - (\sum_{i \geq 1} g^{p^i}),$$

but only the t^{-pc} terms can be eliminated from the negative exponents. Thus we conclude that a degree p Galois extension of $K((t))$ can be uniquely written as the splitting field of a polynomial

$$y^p - y - \sum a_i t^{-i}, \quad \text{where } a_i \in K \text{ and } a_i = 0 \text{ if } p|i.$$

This is in sharp contrast with the characteristic zero case, where there is a unique extension of a given degree if K is algebraically closed.

It is instructive to see what happens with $y^p - y - t^{-1}$ if we try to follow the method of the proof of (1.94). We have to rewrite it as a power series. This is achieved by multiplying by t^p and substituting $y \mapsto yt^{-1}$ to get

$$y^p - t^{p-1}y - t^{p-1} = 0.$$

At the first step we substitute $s = t^{1/p}$ to get $y^p - s^{p(p-1)}y - s^{p(p-1)}$. After $p-1$ blow-ups we get the equation

$$\left(\frac{y}{s^{p-1}} - 1\right)^p - s^{p-1} \left(\frac{y}{s^{p-1}} - 1\right) - s^{p-1} = 0.$$

Thus we keep adjoining the roots t^{1/p^n} , but the equation stays irreducible all the time.

We can also restate our results in terms of Puiseux expansions.

THEOREM 1.96 (Puiseux expansion). *Let $F(x, y) \in K[[x, y]]$ be an irreducible formal power series with $F(0, 0) = 0$. Assume that y^m appears in $F(x, y)$ with nonzero coefficient and m is the smallest such. Assume furthermore that $\text{char } K$ does not divide m .*

Then $F(x, y) = 0$ has a solution of the form

$$y = \sum_{i=1}^{\infty} a_i x^{i/m} \in K[[x^{1/m}]],$$

called the Puiseux expansion of y .

In particular,

$$x \mapsto t^m, y \mapsto \sum_{i=1}^{\infty} a_i t^i \quad \text{give an injection} \quad K[[x, y]]/F(x, y) \hookrightarrow K[[t]],$$

which makes $K[[t]]$ into the normalization of $K[[x, y]]/F(x, y)$. Alternatively,

$\text{Spec}_K K[[t]] \rightarrow \text{Spec}_K K[[x, y]]/(F)$ *given by $(t^m, \sum_{i=1}^{\infty} a_i t^i)$ is the resolution of singularities of the algebroid curve $F(x, y) = 0$.*

Proof. By the Weierstrass preparation theorem (1.88), up to a unit, we can write F in the form

$$y^m + a_{m-1}(x)y^{m-1} + \cdots + a_0(x).$$

By assumption this is an irreducible, and hence $K((x))(y)/K((x))$ is a field extension of degree m . Therefore, by (1.94), $K((x))(y) \cong K((x^{1/m}))$. In other words, y can be written as a power series in $x^{1/m}$.

The normalization of $K[[x]]$ in $K((x^{1/m}))$ is clearly $K[[x^{1/m}]]$, which shows the second assertion. $\square$

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